E Commerce Redesign - case study

Project Overview

Goal: Improve user experience and visual design of "ShopEase," a mid-sized e-commerce app that sells clothing and accessories.

Scope: Focus on UI/UX improvements—navigation, product discovery, checkout flow, and accessibility—without using AI automation or personalization.

1. Problem Statement

User analytics showed:
- *High bounce rate on product pages (45%)
- *Cart abandonment rate at 60%
- *Users complained about confusing filters and cluttered layout
- *Long checkout process (5+ screens)

2. Research

User Research
- *Conducted 14 user interviews (ages 18–40).
- *Identified 3 personas:
  - a.Casual Shopper: wants fast browsing and visual clarity.
  - b.Bargain Hunter: focuses on deals and comparisons.
  - c.Decisive Buyer: values quick checkout and trust indicators.

Key Findings
- Users preferred visual simplicity over dense category menus.
- They wanted clear price visibility and transparent shipping info early.
- Checkout screens felt redundant (too many fields and steps).

3. UX Strategy

Goals
- *Streamline navigation and checkout.
- *Improve visual hierarchy.
- *Build trust with clear, consistent design.
- *Prioritize accessibility and responsive design.

User Flow Optimization

Before:

Home → Categories → Subcategory → Product → Cart → Address → Payment → Review → Confirm

After (Redesigned):

Home → Product → Cart (with address auto-fill) → Payment → Confirm

Reduced from 8 screens → 5 screens.

4. UI Design Improvements

Problem Old Design New UI Solution
- *Cluttered homepage: Too many banners & offers Focused hero banner + clear "Shop by Category" tiles
- *Confusing filters: Text-heavy dropdowns Simplified toggle filters (size, color, price)
- *Inconsistent buttons: Varied shapes/colors Unified primary button color (brand orange)
- *Low trust in checkout: Hidden policies Added secure payment icons, return info upfront

Visual Style

Color palette: Neutral base (white/gray) with accent color for CTAs.

Typography: Used Inter for readability.

Icons: Simplified line icons for clarity.

Accessibility

Minimum contrast ratio 4.5:1.

Larger tap targets (min. 44x44 px).

Screen reader labels for all interactive elements.

5. Prototyping & Testing

- *Low-Fidelity Wireframes
  - Sketches tested for navigation ease.
- *High-Fidelity Prototype
  - Clickable prototype built in Figma.

Usability Testing

8 participants performed 3 tasks (browse, add to cart, checkout).

Results:

Task completion improved from 68% → 95%

Checkout time reduced by 42%

Users rated the redesign 4.6/5 for clarity (previously 3.2/5)

6. Results & Learnings

- *Metric Before After
- *Bounce Rate 45% 27%
- *Cart Abandonment 60% 34%
- *Average Session Duration 3.3 min 4.8 min
- *User Satisfaction 3.2/5 4.6/5

Key Learning:

Thoughtful UX decisions and visual consistency can significantly improve user engagement without relying on AI features.

7. Next Steps

- *Add wishlist and order tracking improvements.
- *Expand testing with larger demographics.
- *Implement dark mode.

E-commerce App UI/UX Case Study (Using Design Thinking Process)

Project: Redesign of "ShopEase" — a mobile e-commerce platform for clothing and lifestyle products.

Objective: Improve usability, navigation, and checkout flow without using AI or automation.

1. Empathize

Goal: Understand users' needs, frustrations, and motivations when shopping online.

Research Methods
- User Interviews: 12 participants (ages 18–40)
- Surveys: 150+ responses on mobile shopping habits
- Usability Audit: Observed how users navigate existing app

Key Insights
- Users get overwhelmed by too many product categories and pop-ups.
- Filters and sorting are confusing; users can't easily find what they want.
- Checkout feels long and repetitive.
- Visual clutter (too many banners and ads) causes frustration.

User Persons
- 1. Aditi (The Busy Professional) — wants fast checkout and minimal steps.
- 2. Ravi (The Deal Hunter) — cares about price visibility and easy comparisons.
- 3. Sneha (The Trend Seeker) — loves visuals, dislikes too much text.

2. Define

Goal: Clearly identify the problem from the user's perspective.

Problem Statement

"Mobile shoppers need a simpler, faster, and visually clean way to browse and buy products, because the current layout and flow make it hard to find items and complete purchases efficiently."

UX Objectives
- Simplify product discovery and filters.
- Reduce checkout time.
- Improve consistency and visual hierarchy.
- Increase user trust with clear UI cues.

3. Ideate

Goal: Generate multiple solutions to the defined problems.

Brainstormed Ideas
- Simplified Navigation: Replace drop-down menus with visual category tiles.
- Sticky Filter Bar: Let users refine products without leaving the page.
- Smart Cart Preview: Show mini-cart at bottom to avoid navigation back.
- One-Page Checkout: Combine address, payment, and confirmation.
- Clean Visual System: Neutral background + single brand accent color.

Wireframing
- Created low-fidelity sketches to visualize layout changes.
- Conducted group feedback sessions with 5 testers for early validation.

4. Prototype

Goal: Create interactive mockups to test the proposed solutions.

Tools Used
- Figma for UI Design
- FigJam for user flow mapping

UI Decisions
- Unified button style for better consistency.
- Product cards with larger images and fewer details (focus on clarity).
- "Quick Buy" CTA added for faster purchases.
- Visual indicators for secure payments and free returns.

Accessibility Considerations
- Minimum 4.51 contrast ratio.
- Legible font (Inter, 16px+ base).
- 44px tap targets for touch interaction.

5. Test

Goal: Validate design effectiveness through usability testing.

Testing Setup
- 8 users performed 3 key tasks:
  - a. Search for an item and apply filters.
  - b. Add product to cart and view cart.
  - c. Complete checkout.

Results

MetricBeforeAfterTask Completion68%95%Average Checkout Time3.5 min2.1 minCart Abandonment60%34%User Satisfaction3.2/54.6/5

Feedback Highlights
- "It feels clean and direct — I know where to click."
- "Checkout finally feels short and simple."
- "Love that there's less distraction from ads."

6. Outcome & Learnings

Impact:
- Engagement up by 2x
- Bounce rate reduced by 40%.
- Checkout flow shortened by 42%.

Key Learning:

Even without AI personalization, human-centered design principles—clarity, consistency, and empathy—create a smoother and more enjoyable shopping experience.

7.User Flow